Which Factors Contribute to the Accuracy and Efficiency of Horse Racing Betting Markets?

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ABSTRACT

This is the abstract.

- A brief introduction to the objectives.
- A brief description of the main work.
- A brief description of the contributions, major findings, results achieved and conclusion.

Acronyms

BHA — British Horseracing Authority
BSP — Betfair Starting Price
CDA — Continuous Double Auction
CLOB — Central Limit Order Book
DPM — Dynamic Pari-Mutuel Market
HKJC — Hong Kong Jockey Club
IP — In-play (after the off)
PNL — Profit and Loss
PP — Pre-play (before the off)
WAP — Weighted Average Price

Glossary

Arbitrage — A risk-free opportunity to make money
Back — A bet that something will happen
Daily Double — A bet on the winner of two consecutive races
Each-way — A combination of a win and place bet

Exacta — A bet on two horses which must come first and second

Lay — A bet that something will not happen

Off — The start of the race

Options Contract — A right but not an obligation to buy or sell a security at a given date.

Overround — The total implied probability of all events in a market

Pandas — Python library for data manipulation

Parlay — A single bet which links multiple consecutive wagers

Place — Come first or second in a race

Prediction Market — A market that facilitates bets on the outcomes of future events

Python — Programming Language

Show — Come first, second or third in a race

Tipster — Professional who publishes betting guides

1. Introduction

As of late, prediction markets have become a hot topic. Services such as Polymarket and Kalshi are offering the opportunity to bet on many world events. There has been research to suggest that prediction markets are more accurate than other forecasting techniques including polls (Berg, 2008). Horse racing markets are also fundamentally prediction markets, and have a long history of widespread use. As a prediction market, horse races provide good data since there are so many and the approximate date and time is known before hand. It is easier to determine their accuracy in this context as there are many repeated similar markets, unlike predictions for political events which can be very far apart.

Racing markets can be considered largely analogous to traditional financial markets, with the only real difference being social convention (Snyder, 1978). They do have one key difference, in that the lifetime of a bet is finite, and the lifetime of a stock is not (Thaler, 1988). In this way, they can be considered similar to binary options contracts, where there are two different payoff values dependant on the result of an external event. The difference between options and wagers here is that the value of an option is dependant on the value of an underlying asset at a future date. Because of this, options are priced using the Black-Scholes model (1973) where it can be shown that buying a combination of the underlying asset and a risk-free investment can produce a payoff equivalent to the option. If not priced this way, an arbitrage is apparent and an inefficiency in the market occurs. Without an underlying asset to hedge against, wagers must be priced differently.

In wider financial markets, it is typical for index funds to be selected from a certain subsection of securities. For example the S&P 500 tracks the top 500 stocks in the United States and the FTSE 100 tracks the top 100 companies on the London Stock Exchange. Both of these are examples of funds limited to investments within a ceratin region. Despite this, little research has been done on how prediction markets can be classified in similar ways for bettors to make informed decisions on selected groups of races only.

1.1. Motivation

I was motivated to research this area due to the recent popularity of prediction markets as a tool for forecasting. As well as this, the regulation surrounding prediction markets is currently heavy within the UK. This is because they fall under the jurisdiction of the Gambling Commission. Many of the newly popular prediction market sites are currently illegal to use within the UK, but established exchanges such as Betfair are completely legal.

Another interesting factor of being classed as gambling means that winnings within the UK are not taxed. This means that an equivalent strategy within a gambling market and the stock market may prove ultimately more profitable in the gambling market as the only fees to be paid are commission to the exchange. In the case of Betfair, commission can be even less than just 2% of successful bets.

I want to understand how different classifications of races can give us new insights into the markets. These insights can be used to aid in the identification of good practices for eliciting forecasting odds, or for the production of profit through betting strategies. Additionally, this research may be applicable outside of the horse racing context, especially in general prediction markets, but also financial markets as a whole.

1.2. Scope

This project explores how the classification of horse races can affect the accuracy and efficiency of the market. The classifications are both external factors and internal market factors. Existing research is used to determine the best practices for identifying accuracy and efficiency. A sample database of horse races will be compiled and tested against to obtain results.

1.3. Aims

This project aims to explore the relevance and impact of external and market factors on the accuracy and efficiency of horse racing markets. Horse races can be classified by various external factors such as race time, location and distance. Internal classifications involve market information such as volume or overround. Similarly, traditional financial instruments can be classified by external factors like their industry or location of operations. They also feature many of the same internal market factors including volume.

1.4. Objectives

To accomplish my aims I will:

- Perform a systematic review of existing research to determine existing standards for market accuracy and efficiency. Identify whether racing markets are generally considered to be accurate or efficient.
- Obtain an appropriate dataset for research.
- Determine the accuracy of my dataset.
- Determine the efficiency of my dataset.
- Determine several classifications of horse races to compare.
- Determine the relative accuracy of the classifications.
- Determine the relative efficiency of each of the classifications.

- Attempt to exploit any inefficiencies for profit.

2. Background

2.1. Bookmaking

2.1.1. Fixed-Price Odds

A fixed-price bookmaker will quote prices (odds) at which they are willing to sell bets. If the bet pays out, the bettor receives the return originally quoted to them. The bettor is given a guarantee of their winnings before placing the bet. It is important to note that it is unlikely a bookmaker will allow a bettor to ‘‘lay’’ a bet, that is to bet that something *will not* happen. Bookmakers make their money by appropriately setting odds such that in all cases (or at least over a period of time), they will ensure a profit. This is done by changing odds to account for the amount of money bet already (balancing the books) and by providing unfair odds that do not sum to one.

There are multiple different ways to represent odds prices. Throughout this report, I will use decimal odds. Decimal odds are given as real numbers greater than one which represent the total payout for a winning bet. For example if a bettor bets £100 dollars at odds of 2, their winning payout would be £200 - a profit of £100 or 100%. The payout for all losing bets is 0 (a loss of all money bet). Assuming that the expected value of a wager is zero (no profit or loss), the implied probability of winning is given by $1 smallover odds$.

2.1.2. Parimutuel Odds

Parimutuel betting (commonly known as a ‘tote’ in the UK) is a system in which money for all bets is placed into a pool and then distributed to winning bettors. This means that bettors are not given an exact payout quote at the time of purchasing their bet, but they may be given an estimate. Tote operators make money by taking a certain percentage of the pool for themselves, typically called the ‘‘track take’’. These types of bets are typically facilitated by the track operator themselves, such as the HKJC (Hong Kong Jockey Club, n.d.).

2.1.3. Betfair Exchange

Betfair is a UK based company which operates a betting exchange service. A betting exchange is different from a typical bookmaker in that instead of the operator quoting odds prices or offering a tote service, it allows users to form bets with one another. Betfair does this using a CLOB, allowing users to either back (long) a bet or lay (short) one. Within the context of a horse race, this means that the win market (where bets are paid out only if the horse places first) features a different asset for every selection (runner) in the race. They can be thought of as largely analogous to options contracts, where the layer agrees to pay the backer a certain sum of money dependent on the outcome of the race.

A unique feature of the Betfair racing market is its "starting price" betting. Bettors who use the starting price option have their bets matched into the market at the off (when the race begins). They are also able to set minimum odds to back or maximum odds to lay. The price determination maximises volume and gives backers and layers the same price. In order to do that, existing positions within the CLOB may be matched against as well. Starting price bets aren’t guaranteed to be matched if the odds limit is at or above the starting price (Betfair, n.d.).

2.2. Literature Review

2.2.1. Accuracy of Horse Racing Markets

2.2.1.1. Measurement of Forecast Accuracy

One of the earliest methods for scoring the accuracy of a forecast is provided by Brier (1950). The Brier score is determined as a result of many predictions using the mean squared difference between the estimated probability and the actual outcome. Formally:

$$BS = \frac{1}{N}\sum_{t=1}^{N}(f_t - o_t)^2$$

where f_t is the forecast probability, o_t is the actual outcome (0 if did not happen or 1 if did) and N is the number of forecasts. In this system, a perfect forecast is given a score of zero. This can only be achieved if the forecast guesses with 100% certainty every time. Ranked Probability Score (Epstein, 1969) is an extension of the Brier score to support multiple outcomes Ranked Probability Score with binary outcomes can be shown to be equivalent to the Brier score (Weigel, 2007). Brier score will be the most useful in this report, as in (most) betting markets there are only two possible outcomes: a winning bet or a losing bet.

Good (1952) developed a method for scoring estimates of probabilities now known as log-loss. This method has since become popular as a scoring mechanism for machine learning models as it highly punishes overconfidence in incorrect predictions. For a binary outcome:

$$LL = -\frac{1}{N}\sum_{t=1}^{N} o_t \log(f_t) + (1 - o_t)\log(1 - f_t)$$

where f_t is the forecast probability, o_t is the actual outcome (0 if did not happen or 1 if did) and N is the number of forecasts. Log-loss has become incredibly popular as a cost function (function that scores prediction accuracy) for machine learning. Machine learning textbooks (Goodfellow, 2016; Murphy, 2022) share that log-loss should be considered the standard cost function. Since log-loss easily supports binary outcomes, it can be applied in addition to Brier score as a measurement of accuracy for betting predictions. Both Brier scores and log-loss will be useful in ranking the accuracy of markets. The primary difference between them is how harshly they rank confident incorrect guesses.

Skill scores are a way to measure forecast skill relative to a given baseline (Murphy, 1988). These can be applied to both the Brier score and log-loss.

$$SS = 1 - \frac{score}{score_{reference}}$$

A skill score of zero indicates that the forecast is just as good as the baseline forecast. Conversely to the normal Brier score and log-loss, a higher skill score is an indication of better performance. Negative skill scores are worse than the reference and positive skill scores are better than the reference.

Wunderlich and Memmert (2020) demonstrated that the betting returns of a strategy are not necessarily indicative of accuracy in a forecasting model, despite demonstrating profitability. Instead, there is a difference in a model forecasting accurately and profitably. Later in this report, I will explain and demonstrate some betting strategies, however they are only relevant to the efficiency of the market. This is a prime example of what Wunderlich and Memmert explain. The

strategies can be successful, even without predicting anything to do with horse racing.

2.2.1.2. Use of Markets for Forecasting

Current research shows prediction markets to provide accurate estimates of outcomes for elections over both long and short time frames (Berg, 2008). Rothschild (2009) showed that, when used appropriately, prediction markets can perform significantly better than the very best polls. Elections are just one example, but prediction markets have been used to great success across many different domains of forecasting. One study found prediction markets to outperform a survey when used to predict the reproducibility of studies published in psychology journals (Dreber, 2015).

Because of the demonstrated favourite-longshot bias (see more below) in horse racing, many studies attempt to convert odds given by bookmakers or markets into ‘‘real’’ probabilities. Štrumbelj (2014) used Shin’s (1993) model to create more accurate forecasts than those created through odds normalisation or regression. These methods are criticised by Clarke et al. (2017) for potentially producing negative probabilities and in the case of normalisation, not accounting for the favourite-longshot bias. They share that an alternative power method instead performs as well as Shin’s, or better. The necessity of these models in the first place demonstrates the lack of accuracy present within the markets. If the given odds can be altered using some function to provide forecasts of greater accuracy, then the market obviously isn’t producing the best possible forecasts inherently.

Strijbis and Arnesen (2019) find that the setup of the market influences accuracy the most, while features of the event and the market’s participants have an insignificant impact. Market setup incorporates factors such as liquidity and the market mechanism. Prediction markets require the right incentives in order to produce the most accurate and up to date forecasts (Chen, 2010). Markets will also suffer if a group of insiders are able to influence the outcome. If the goal of the market is to produce the best forecast, then it must incentivise immediate truth-telling instead of deception. A typical parimutuel racing market fundamentally does not meet this requirement. This is because market participants are encouraged to wait until the very last moment to bet, so as to hide their beliefs about the probability of a win. A CDA (the primary mechanism of the Betfair Exchange) encourages price discovery over time but may still incentivise participants to to lie about their true valuation. Since the price of a parimutuel market is not set until the market closes, there is no incentive for the market to track changes in information unlike a CDA.

Pennock (2004) suggests an alternative ‘‘dynamic pari-mutuel market’’ (DPM) which functions similarly to typical parimutuel betting. A DPM allows investors to buy and sell bets at a changing rate determined by the market, giving them an opportunity to hedge or sell their bets early.

2.2.1.3. Favourite-Longshot Bias

The favourite longshot-bias has been demonstrated within racing markets as far back as 1949 (Griffith, 1949). This is a phenomenon in which bettors tend to prefer to bet on outlier horses, as opposed to the safer options. In turn, this moves the odds given by the market as favourite horses are given worse implied probabilities of winning and longshots are given better implied probabilities of winning. Other than the long and short-odded horses, Griffith found that the odds for horses were on average accurate.

Some studies have attempted to determine why this is the case. Weitzman (1965) used utility theory to derive a utility curve for one bettor and found it to be similar to the theoretical one proposed by Markowitz (1952). To follow this study, Ali (1977) observed over 20,000 races and found the favourite-longshot bias to be present. Ali creates a wealth utility function and determines that the average bettor values risk, thereby causing the discrepancy in prices. Subsequent research has also conjectured that bettors have a preference for risk at the racetrack. Kanto et al. (1992) found this to be the case using a data sample from Finland.

Golec and Tamarkin (1998) suggested an alternative explanation for the favourite-longshot bias. They claimed that it was possible that bettors could actually be risk-averse and instead have a preference for skewness. Skewness here is the asymmetry of betting returns about the mean. To back this up they examined popular betting strategies and determined that they were better aligned with a preference for skewness rather than risk. Another theory is put forward by Snowberg and Wolfers (2010) which states that the bias can be accounted for by the misperceptions of risk by the bettors.

The bias has also been shown to not be universal. In contrast to the evidence found in North American markets, Busche and Hall (1988) found that bettors at the Hong Kong Jockey Club do not exhibit the same apparent risk-loving behaviour. They posit that the larger betting volume in Hong Kong may influence the bias.

2.2.2. Efficiency of Horse Racing Markets

2.2.2.1. Measurement of Market Efficiency

Fama (1970) set out three tests of efficiency for markets depending on how rigorously they can be described as efficient. An efficient market is one that is said to fully incorporate all the available information into the prices. This means that, for an efficient market, it should be impossible to create a profit generating strategy. For a market to be *weak form* efficient, the information set is constrained to that of historical price data. The *semi-strong form* tests also incorporate information widely available to the general public. In the *strong form* tests, the insight of investors with insider knowledge is also considered.

A significant amount of research into the study of efficiency in racing markets is specifically focussed on parimutuel markets. This is the most common market for racing in the United States as well as Hong Kong. Comparitively little exists for markets using a CDA such as Betfair. This makes this report particularly interesting as a way to compare parimutuel markets with a CDA. It also means that the work done here is potentially has greater applicability to external financial markets, which typically use a form of CDA as their market mechanism.

2.2.2.2. Relationship of Exotic Bets

A lack of market efficiency can be demonstrated by comparing the prices of exotic betting options with equivalently produced parlays. Exotic bets involve betting on a combination of different outcomes for different horses within one wager. Popular examples include the daily double, in which the first two winning horses of the day must be predicted correctly, and an exacta, where the first two horses of a race must be predicted in order. A parlay is similarly a wager of multiple outcomes, is constructed from multiple separate bets with the winnings from earlier bets accumulated into future ones.

Certain exotic bets can be shown to have an equivalent payoff function to an appropriately constructed parlay. For the market to be considered efficient, both the exotic bet and the parlay must have an equivalent price. This difference in a typical financial market may be considered an arbitrage opportunity as one would be able to buy at the low price and sell at the high price, guaranteeing a profit. Within parimutuel markets however this is not possible, as bets cannot be bought and sold at different prices. This is fundamentally similar to established options pricing theory. The payoff of an option can be shown to be replicated by purchasing a quantity of the underlying asset and a risk free asset (Black, 1973; Cox, 1979).

Ali's (1979) analysis of 1,089 racing days uses this principle to compare the prices of daily doubles and their equivalently constructed parlays. In this analysis, Ali finds that the efficient market hypothesis could possibly hold for racing markets as the difference in price between the exotic bets and the parlays is not statistically significant. In a different study however, Asch and Quandt (1987) find that for both exacta and daily double bets, their payoffs are not the same as equivalent bets on individual horses, thus demonstrating market inefficiency.

It is uncommon for betting exchanges to offer any betting options for horse races outside of win, place and each-way. This means that a comparison of exotic bets and parlays is not possible. While it may be possible to compare the prices of the win and place bets to the each-way bets, the limited volume of non-win markets makes this difficult.

2.2.2.3. Betting Strategies

The most obvious way to demonstrate a lack of market efficiency is to produce a simple, profitable betting strategy.

In a study of 1,730 races in American parimutuel win markets, Snyder (1978) showed that because of the favourite-longshot bias, prices were indeed more favourable for the favourite horses, but the track take was too significant for a strategy of favourite betting to be profitable. This indicates some level of *weak form* efficiency, however without the track take present (and betting habits remaining the same), it would certainly be impossible to describe the market as informationally efficient. Snyder also demonstrated that this bias exists among betting experts (or ''tipsters'') which makes it seem unlikely that the market is *strong form* efficient.

Hausch et al. (1981) set about creating a technical betting system for place and show markets. Their strategy only used historical pricing data and so attempts to demonstrate a lack of *weak form* efficiency. They argued that their system successfully exploited a market inefficiency and was able to generate a consistent profit. To do this, they derived estimated place and show probabilities from the implied win probabilities given by the win market using Harville's (1973) method. The win market is assumed to be informationally efficient. This allows them to profit through mispricing of show and place odds relative to the win market. With a follow up paper, Hausch and Ziemba (1985) greatly improved the system making it more profitable and applicable to any racetrack. They also demonstrated how more people using the system would ultimately cause the market to become efficient.

3. Methodology

3.1. Data Source

Historical odds pricing data is typically very difficult to find if not impossible from most bookmakers. This was a significant factor in choosing to use prices from the Betfair exchange.

Betfair provides historical betting data at multiple levels of resolution including a generous free tier. The data provided in this set is structured like exchange data, with multiple entries per race documenting changes in prices over time. For my purposes, this is beyond what I need as I am only interest in the BSP and not the price evolution. I used this dataset at first but found this is beyond what I need for my purposes, as I am only interested in the BSP and not the price evolution.

Betfair also provides a directory of starting price files and allows users to filter them by location and market type. This dataset contains some useful information not included in the free tier of the standard historical data files such as trade volume and minimum and maximum prices for each selection. It is also significantly larger, containing data from as far back as 2008, compared to 2016 with the other set. In total, the dataset contains odds data for over 170,000 win markets in the UK. This is the dataset I chose to use as it was the most comprehensive and most simple to use. It is worth noting that the market mechanism for the Betfair exchange is different from the typical parimutuel markets explored in other research and so results using this dataset can not be directly compared, but they are still worth using to explore the impact of a different market type.

I had originally also considered using multiple datasets to compare different markets. To this end, I obtained a dataset from the Hong Kong Jockey Club but it was not sufficient for this project as it only contained information from 2,000 races.

3.2. Classification

I decided on a few different classifications to compare between. These were decided mostly due to what was available within my dataset. I wanted to get a varied range of groupings as it was not clear which, if any would produce interesting results. For external factors, I chose: race number, distance, hour of day, race type, location, number of runners. Number of runners could also potentially be considered a market factor, as it affects the number of possible selections to choose from within the market. The market factors are pre-play trade volume and overround.

3.3. Accuracy

As a baseline, the existence (or non-existence) of the favourite-longshot bias within the dataset should be established. This is important as it is an expected market inaccuracy consistently found across the literature. A change in the bias could indicate different sub groups of bettors within horse racing with different utility estimates. To look for this, I will plot the implied probabilities given by the odds against the true (observed) probabilities. This will be a clear indicator of accuracy, as these two values should be the same for an accurate market.

Brier scores and log-loss will also be used. With the entire dataset given as a benchmark, the groups produced by the different classifications will be compared to see if they are to be considered more or less accurate by these test statistics. These values give a more concrete value to compare accuracy between.

3.4. Efficiency

Due to my choice of dataset, I do not have an opportunity to compare prices of exotic bets to parlays. This means that the most straight forward way to test for market efficiency is to subject the market to various betting strategies. The existing research for this is done in the context

of parimutuel betting, where it is known how exactly a bet will change the odds within the market (Hausch, 1981; Hausch, 1985). Because of this, I will not be able to employ the same strategies, beyond the simplest of betting the same amount on favourite horses.

The first strategy I will implement simply bets one unit on the favourite horse. This is one of the simplest and most widely implemented strategies. As a slight twist on this, I will implement a similar strategy where the size of the bet is determined by the implied probability of the odds price. This is intended to hedge potential losses from horses with a lesser chance of winning.

I will also test a slightly more involved strategy. This strategy will take a running average of how likely it is that the favourite horse has won and take this value to be the true probability of the favourite horse winning. The probability will be plugged into the Kelly Criterion in order to determine the size of a bet. That is: $f* = (bp - q)/b$ where $f*$ is the fraction of wealth to bet, p is the likelihood of the bet winning, q is $1 - p$ and b is the proportion of wealth won if successful (Thorp, 2006). While this strategy will help identify market inefficiencies, it is not pratical for real use as it does not take into account how placing a bet will move the market. Since this strategy relies only on historical price data, it intends to demonstrate a lack of *weak form* efficiency.

4. Implementation

4.1. Data Acquisition

I produced a Python script to automatically download all of the files found in my dataset and process them to be used within Pandas. The script saves the data to binary files for quick retrieval. It makes sure to not redownload files it has already obtained so it can easily update the existing database. As well as this, it automatically slows down when rate limited to ensure that all files are downloaded.

Another script transforms the flat file structure into a relational one. The relational structure avoids duplicating data and keeps unrelated information separate. This data is formatted into three different tables: events, races and, runners.

4.1.1. Events

Field	Data Type	Notes
`event_id`	`Integer`	Unique identifier provided by Betfair
`location`	`String`	
`event_date`	`Datetime`	

4.1.2. Races

Field	Data Type	Notes
race_id	Integer	Unique identifier provided by Betfair
race_type	String	Indicates the format of the race: handicap, hurdles, chase, etc.
race_time	Datetime	
distance_f	Integer	Furlongs are one eigth of a mile long.
race_no	Integer	Races are given numbers indicating their sequence position within an event.
event_id	Integer	Relational key.

4.1.3. Runners

Field	Data Type	Notes
selection_id	Integer	This identifier is unique per horse, not per entry.
selection_name	String	
win_lose	Boolean	Win=True, Lose=False
bsp	Float64	Betfair Starting Price
implied_odds	Float64	1/BSP
ppwap	Float32	Pre-play weighted average price.
morningwap	Float32	Weighted average price of bets before 11AM GMT
ppmax	Float32	Pre-play max price.
ppmin	Float32	Pre-play min price.
ipmax	Float32	In-play max price.
ipmin	Float32	In-play min price.
morningtradedvol	Float32	Trade volume before 11AM
pptradedvol	Float32	Trade volume before the off
iptradedvol	Float32	Trade volume after the off
race_id	Integer	Relational key.

4.2. Software Development

I decided to use Python through Jupyter Lab to write and run my software as it is what I had the most experience with for statistical analysis. Python also features extensive libraries for data analysis and statistics. Particularly useful in this project is the Pandas library, which provides a significant amount of functionality for working with structured data.

4.2.1. Data Structure and Statistics

4.2.1.1. Pre-Processing

I pre-processed my data to include some more fields that would be useful. Pre-processing fields like this means that the calculations do not need to be repeated later on.

The first field to be added was a normalised probability field. This is useful as the implied odds given by the starting price (almost) never sum to be one. The sum of implied odds given is known as the ‘‘overround’’. An overround greater than one for a back market shows that there is negative arbitrage and a weighted selection of back bets is guaranteed to lose money. Since the data for this project is using the BSP, it can trivially be shown that it is possible to arbitrage almost every single market. This is because for BSP, both back bettors and lay bettors are given the same price. In real terms this arbitrage is actually impossible to achieve however, as the BSP is not possible to know before the off, and by betting this way you are certain to move the market.

I then added fields for partial pre calculation of the Brier score and Log-Loss. This is particularly useful as it means that to determine the scores for a certain subset of races, they can simply be summated using the groups generated in the next section.

4.2.1.2. Categorisation

Given the available data from my dataset I decided upon a list of ways to categorise the horse races: race number, distance, hour of day, pre-play trade volume, race type (handicap, hurdles, etc.), location, number of runners and, overround. To do this I produced a list containing these different classifications of races created using the `groupby` method in Pandas. The `groupby` method makes it easy to perform aggregate calculations per group created through the method. This means that a function such as `sum` will be ran per group, instead of for all data in the DataFrame.

4.2.2. Trading Strategies

Trading strategies are implemented using a system in which the strategy function receives a list of available runners to bet on and returns a ‘‘bet slip’’. The bet slip contains information about which race and runner it chose, as well as the amount bet and whether or not the selection won. The amount bet is able to be submitted as either an absolute value, or as a percentage of the currently held bank value.

The bet slips are passed to a function which actions the bets and determines the PNL for each one. It also sorts the bets over time to generate a timeline of events.

Each betting strategy is written using vector based functions. This allows the underlying library (Pandas or Numpy) to perform many operations in parallel. There are two primary benefits to this. Firstly, it significantly improves the readability of the code as the need for looping is removed. The code is much more representative of what is actually happening. Secondly, there is a major performance increase by writing the software in this way.

4.2.2.1. Naive Strategies

4.2.2.2. Learning Strategy

4.2.3. Graphing

To plot my results, I used the program Gnuplot. Python comes with its own plotting utilities (in the form of Matplotlib), however I was not satisfied with their capabilities as I found it awkward to position elements to my liking. It was also difficult to produce a format easily embeddable within my document preparation system (Groff). The workflow of producing output datafiles in Python and then producing plots through Gnuplot in Jupyter was extremely

straightforward.

5. Results, Analysis & Evaluation

5.1. Initial Observations

For most of the groupings, trade volume follows the number of races as can be seen in Figure 1. This is of course different for the categorisations of volume (per race) and overround, which have been specifically broken up into 10 sections of equal size.

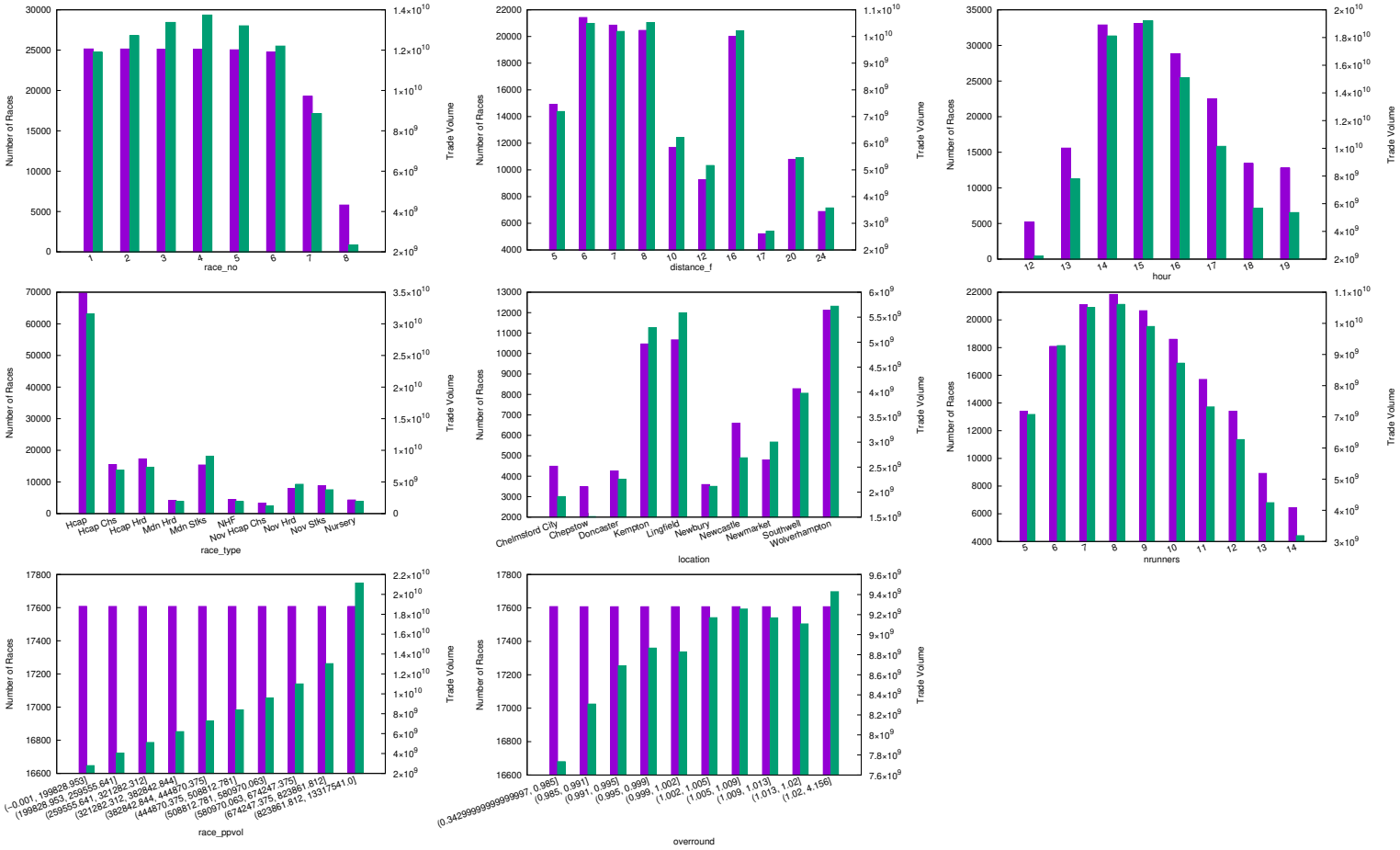


Figure 1: Number of Races and Trade Volume per Category

For the entire dataset, the volume traded appears to be roughly normally distributed about the mean implied probability of 0.5. The number of selections however is positively skew, with significantly more selections being given a lower implied chance of winning. This shows that most horses will be considered unlikely to win a race. More money is traded on a smaller proportion of horses with greater chances of winning.

An interesting observation are the somewhat frequent spikes in prices. These are all observed at the points in which the market tick size changes (BetexTrader, n.d.). The tick size is the difference between prices that are allowed to be offered. I have used the tick size to group selections based on their BSP, as the reported BSP is not the same as the actual price matched against. The tick sizes are used to adjust the frequency values in the plots.

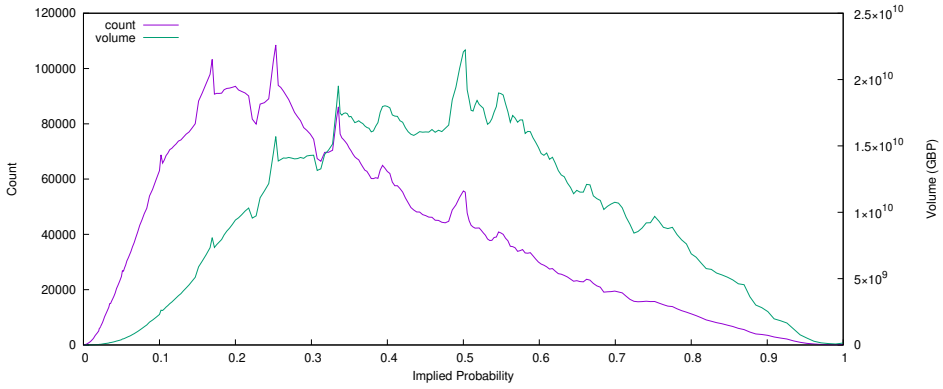


Figure 2: Frequency of Count and Volume of BSP Values Across All Races

The volumes per grouping tell a largely similar story. For each different grouping, the distributions are typically the same as per the whole dataset, except scaled by quantity. Distance, hour, location, and number of runners all fit this trend.

Race number becomes more positively skew as it increases. This shows that over the course of the racing day, bettors are more likely to trade on horses with a lower implied chance of winning. We can see this is not simply due to the odds being offered as the frequency plot remains almost exactly the same across all races.

The volume for race type is another outlier. Handicap races are massively positively skew while others tend to be negatively skew. Handicap races are by far the largest race type in the data set and so have a significant impact on the overall weighting. It could be the case that on a dataset without featuring handicap markets, volumes generally tend to be negatively skew, instead of symmetric.

For both total pre-play trade volume and overround, the volume of the market tends to shift towards being negatively skew as they increase. This trend is particularly pronounced for pre-play trade volume. As total trade volume increases, trade tends to center around horses with a greater chance of winning. It is so extreme that in the highest bracket of volume, well over 50% of trades are being made on selections with an implied probability of winning of over 50%. I believe that this is signifying an increase in professional trading activity, where traders try to scalp prices on popular horses for bigger races.

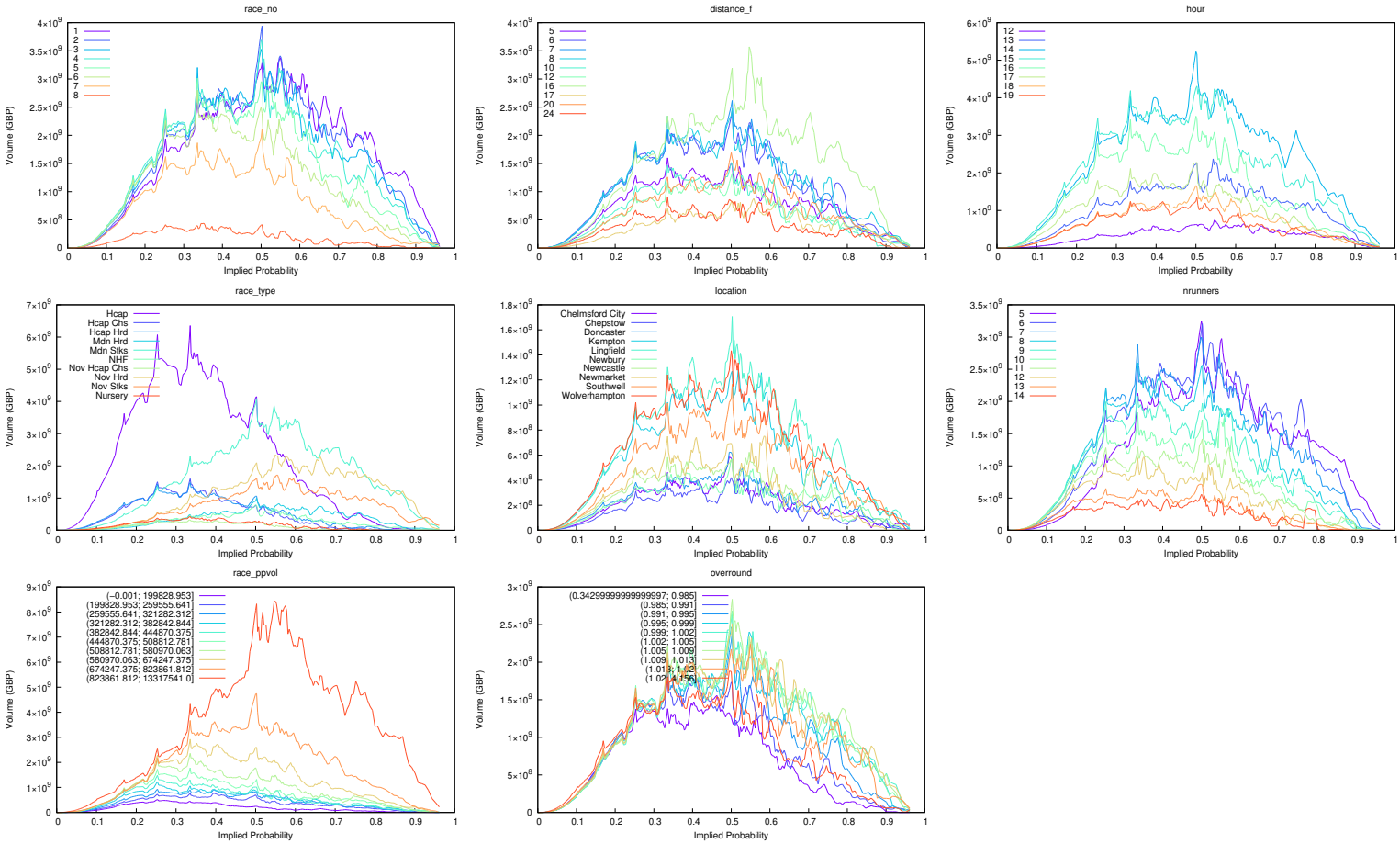


Figure 3: Frequency of Volume for BSP Values by Category

Frequency follows the same trend for all categories except for pre-play trade volume. All other groups are positively skew, being scaled relative to their group size. For trade volume however, the frequency becomes more negatively skew as the total volume increases. This is similar to the previous observation for trade volume. As pre-play trade volume increases, the distribution of probabilities assigned to horses changes as well. Prices going down due is indicative of greater market condfidence in the favourite horses.

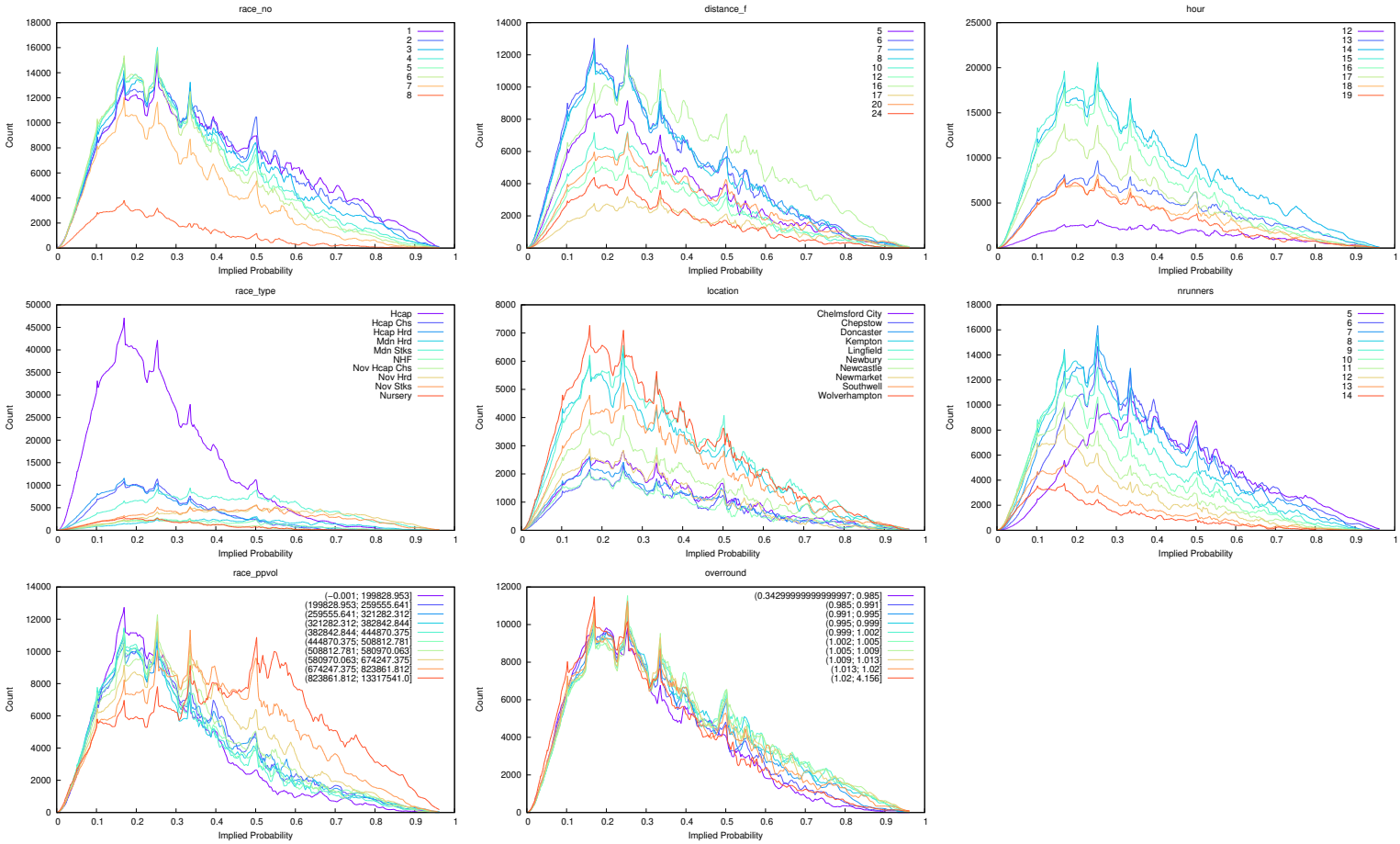


Figure 4: Frequency of Count for BSP Values by Category

5.2. Accuracy

5.2.1. Favourite-Longshot Bias

If the favourite-longshot bias is present within this sample, I would expect to find that favourites win more often than implied by their given betting odds. Similarly, longshots should win less often than implied by their betting odds. From research done in American markets (Griffith, 1949; Ali, 1977), I am expecting to find the favourite-longshot bias within my sample.

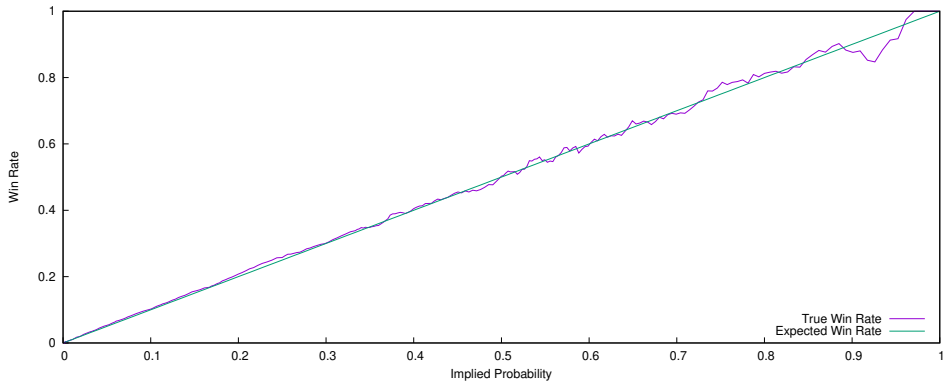


Figure 5: Win Rate Compared to BSP For All Races

When observing the entire dataset, it is clear that the favourite-longshot bias is not present. The mean win rate closely tracks the expected win rate at all implied probabilities. If the favourite-longshot bias was present, the mean win rate would appear greater than the expected win rate for low implied probabilities and lesser than for high implied probabilities. Variance within the higher expected win rates can be largely explained by them making up a representatively small part of the sample.

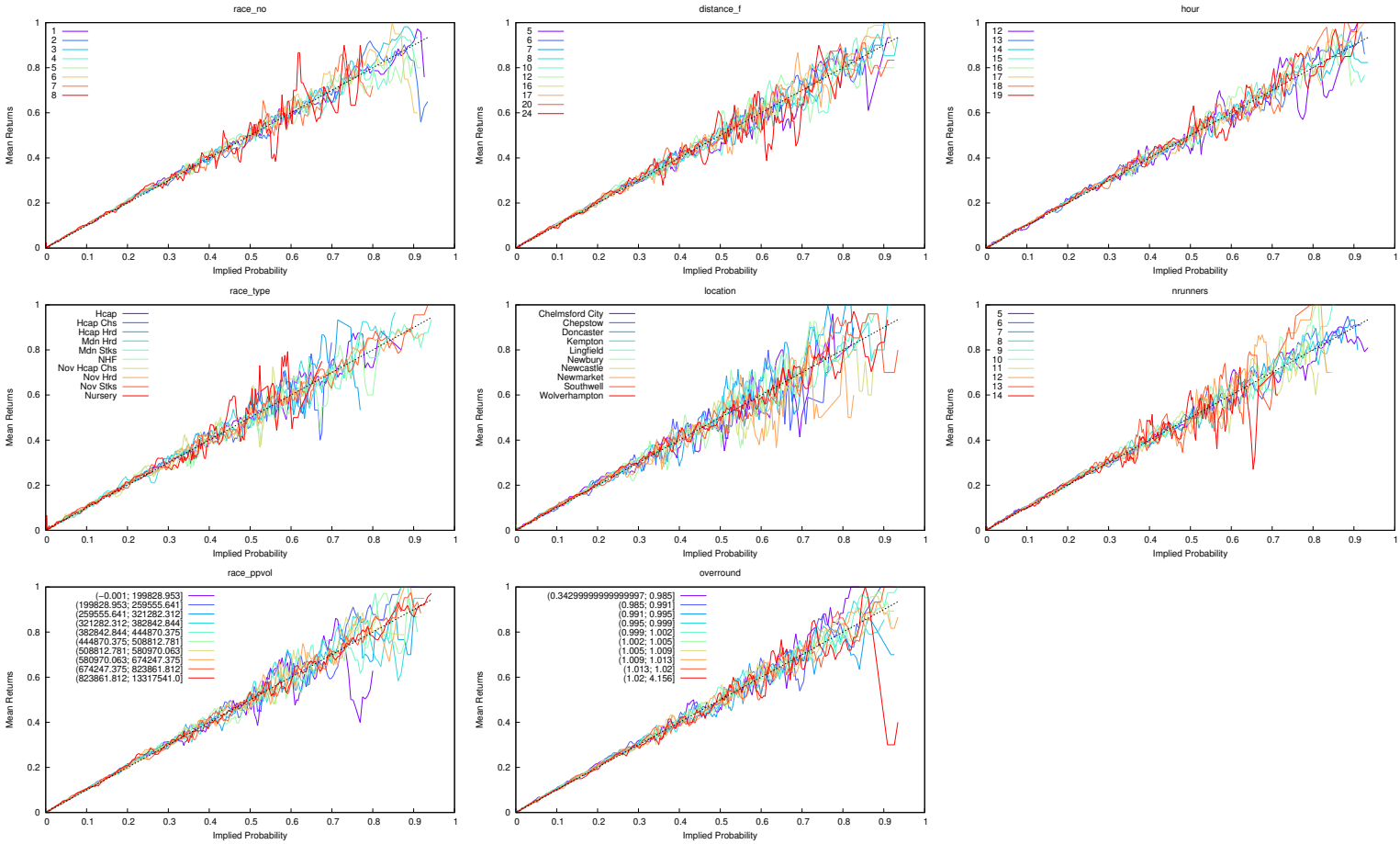


Figure 6: Win Rate Compared to Implied Probability by Category

Similarly, for all categories it appears there is no sign of the favourite-longshot bias. They all have greater variance, but this is simply due to a smaller sample size.

This is a stark contrast from most previous research, with the only other example I know of a racing market not exhibiting this bias is the Hong Kong Jockey Club (Busche, 1988). I expect the primary reason for this difference to be the use of a CDA for price discovery, rather than a parimutuel system. The CDA is encourages greater price discovery and its format is familiar to those with experience in financial markets. This could suggest that the composition of the Betfair Exchange market leaning further towards groups of insiders or professional bettors, rather than normal punters.

If this is the case, then there may well exist a favourite-longshot bias in the odds provided by normal UK sportbooks or tote operators. Typically, this would be an arbitrage opportunity as the two markets are offering differing prices on the same outcome. There are however a few points that negate any usefulness of this arbitrage for someone attempting to exploit it. First of all, traditional sportsbooks do not allow a bettor to lay a bet which is typically necessary for an arbitrage opportunity. You could instead bet on all other winning outcomes to replicate the lay bet, but this will encounter the next problem. Bookmaker overround is probably far too significant to overcome, requiring the bettor to have an edge of well over 20% to actually make any money. Betting on multiple selections causes the impact from overround to accumulate. The

most signficant issue is not to do with the numbers, instead it is the fact that almost all bookmakers will simply ban you from using their service for attempting to do this.

5.2.2. Forecast Test Statistics

To determine the test statistics, the statistic is determined first for each individual race and then averaged across the whole grouping. This is done as we are less interested in the score as determined per runner as we are per race. This is because each runner is not an isolated event, and is instead packaged within a race. Organising by race prevents extra external factors from affecting the results. For the data set as a whole, the Brier score is determined to be 0.0951 and the log-loss is 0.3167. Skill scores can be used to show that both of these metrics are better than randomly guessing the likelihood of a horse winning a race.

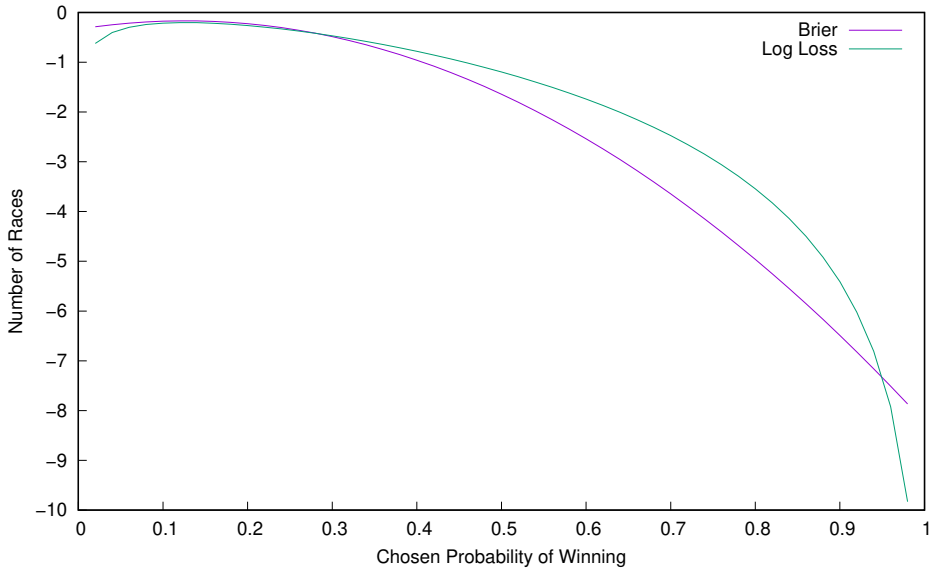


Figure 7: Skill Scores of Random Predictions Compared to Market Predictions

Figure 7 compares different rates of estimating the probability of a horse winning. At 1, the model predicts every horse to win every race with a 100% chance and at 0 it predicts every horse to not win with a 100% chance. The comparison makes the differences between Brier and log-loss extremely clear, as log-loss drops sharply at higher estimated probabilities due to its greater punishment of incorrect guesses. Both of these scores show that the market performs better than random.

I would expect the skill scores to become greater for groups of larger sizes. This is because the larger pool would provide more data for for bettors to use to make predictions with. The sizes of the groups can be found in Figure 8. It would also make sense for some particularly small groups to have outlying skill scores as they will be more susceptible to variance.

To identify accuracy within the different groupings, brier and log-loss are calculated the same way as before, accounting for selections within races.

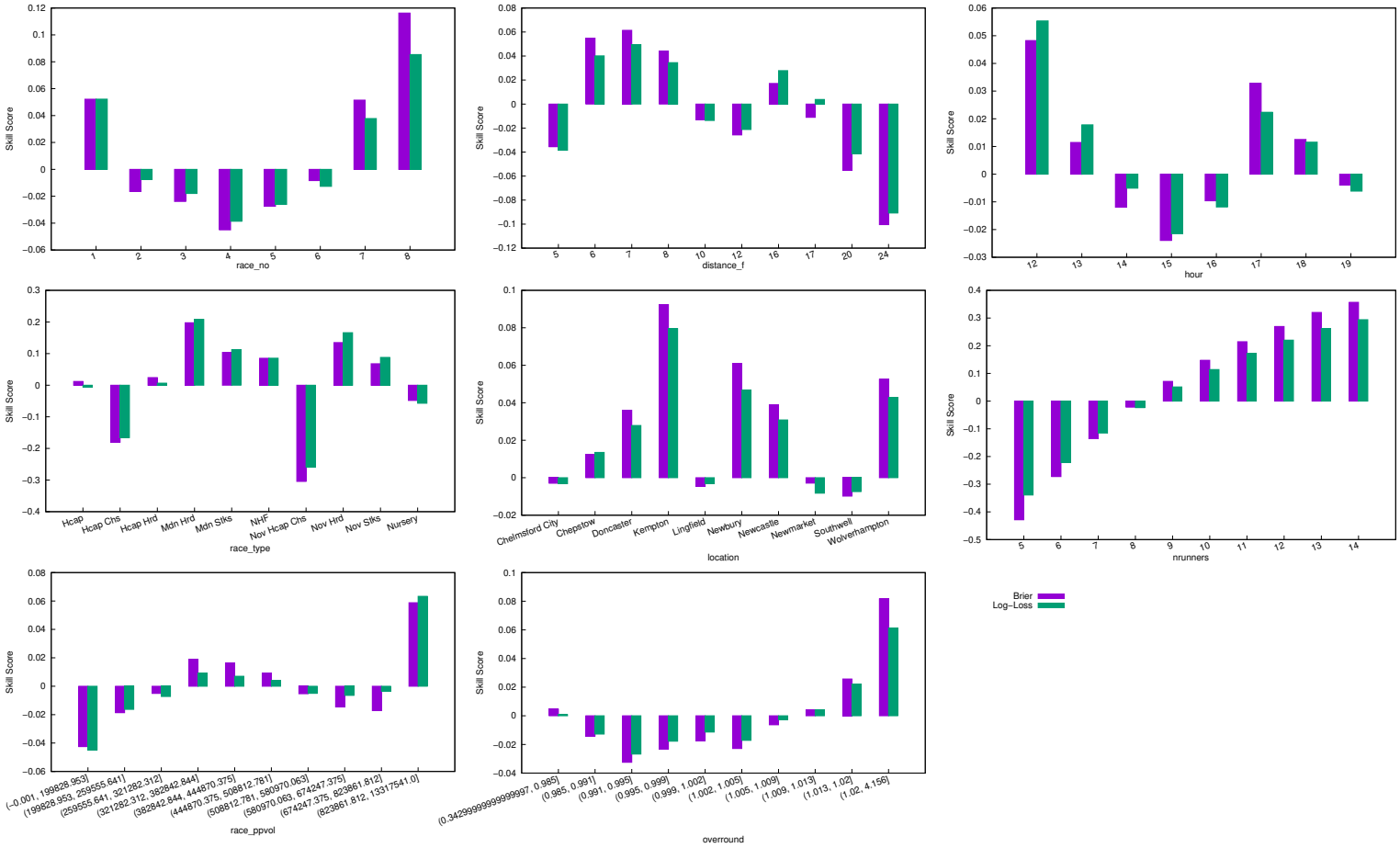


Figure 8: Brier and Log-loss Skill Scores by Category

Throughout every category, both skill scores are largely similar. Figure 8 compares the sill scores for every category. There are a few occasions where the scores conflict and one is negative while the other positive. In all instances of this, the skill scores are both very close to zero. The difference will be due to slight differences in the the distribution of skill scores as seen in Figure 6.

Some skill scores appear to be correlated with the size of the categories or the total trade volume. (Figure 1). Categories that do appear correlated are: race number, distance, hour, and location. Race number appears to have the skill scores negatively correlated with volume, as does hour. However, distance and location have skill scores that positively correlate with volume. The contrast between these categories shows that volume can not necessarily be used as an indicator of market accuracy.

Other categories appear to have skill scores unrelated to the relative volume or race count. The prime example is number of runners; skill scores increase as the number of runners does. This behaviour is to be expected because of our test statistics. A greater number of selections in a race means that the average probability estimate will decrease and the number of non winning horses increases. Number of runners has the greatest range of skill score values and appears to be the most significant factor in determining the accuracy of the markets by these metrics.

Race trade volume and overround both exhibit a behaviour of having low or negative skill score before sharply rising at the high end of the categories. Overround is a representation of the total probability of prices given in the market. For an overround of 1, all prices add up to a total probability of one event occuring being 100%. Odds offered by traditional bookmakers typically have an overround much greater than one, this is the margin that they make. Because of this, I would have expected the accuracy for overround to peak around 1, as at higher and lower values there are more prices which are explicitly wrong. The results here contradict that expectation. Trade volume per race is mostly a measure of race popularity, and so increased accuracy for the greater trade volumes makes sense. There is more money in the market and probably a greater proportion of informed bettors.

The only category not mentioned thus far is race type. There seems to be no relationship between the race type and accuracy. Size and volume of the groups does not impact the accuracy here, where handicap races (by and far the most popular) have a similar score to handicap hurdles or nursery races.

5.3. Efficiency

Three betting strategies were trialled in an attempt to exploit inefficiency within the market. The first two strategies bet on the favourite horse for every race. The first one simply bets one unit on every favourite, while the second bets the inverse of the starting price. The second strategy makes a weak attempt to ‘‘hedge’’ by betting lower when it is apparent that the probability of winning the bet is lower. The third strategy is adapted from the technique used by Hausch et al. (1981).

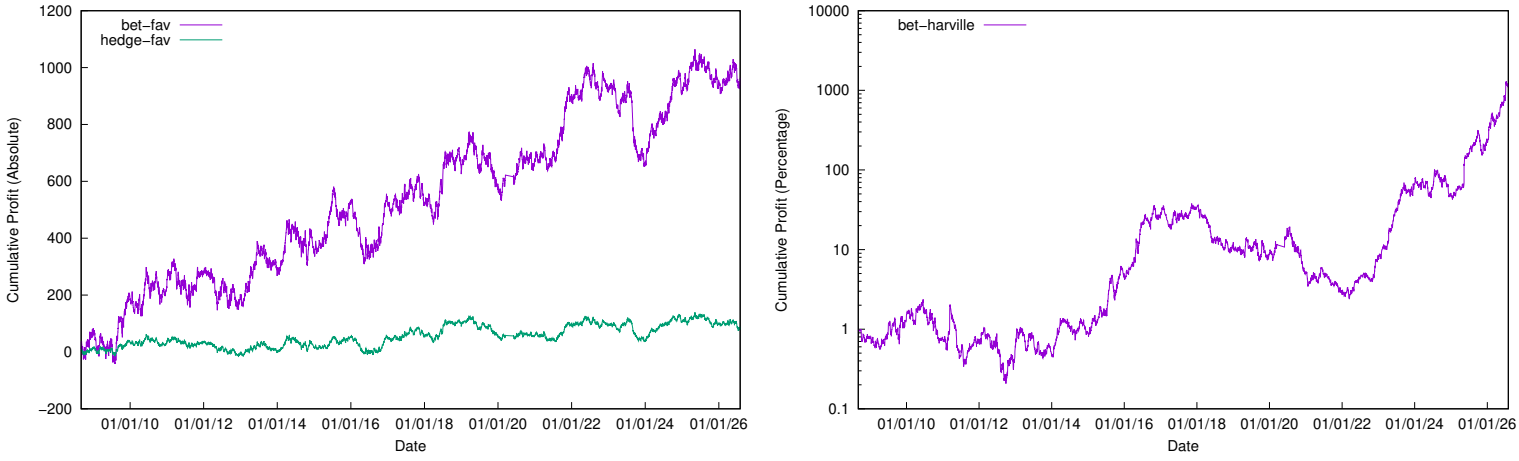


Figure 9: Results of Betting Strategies Applied to All Races

The results of these strategies when applied to all races can be seen in Figure 9. When hedging, the resulting profit stays exactly at zero. This is to be expected in an efficient market as the expected value of the profit of any bet hedged this way should be zero. The favourite betting strategy at first glance appears to actually provide a profitable system. However, the profit here is actually realtively small compared to the number of wagers. Given that over 170,000 bets have been placed, the final result is a return of less than 1%. While technically a profitable strategy, it is essentially worthless due to the low returns. A return this low is not substantial enough to claim a lack of efficiency in the market.

The more advanced strategy looks promising from the outset. Over the time period, it is able to generate a total profit over over 1000 times the initial wealth of the bettor. This of course is not representative of the results that would truly be earned, as a bettor will not be able to place large bets at favourable prices without moving the market to an unfavourable position. The Sharpe ratio (Sharpe, 1994) can be applied to establish a metric of strategy performance. For the entire dataset, the annualised Sharpe ratio of the more advanced strategy is 0.9679. This is generally considered to be a somewhat weak score. Investors typically believe a Sharpe ratio greater than 1 to be adequate and greater than 2 to be strong. The reason the score is so low is because the variance of the returns is too great, and so there is a lot of risk in the investment. While it seems to perform well in the test here, it is likely to have significant downward periods (as exhibited in Figure 9) and so is unappealing.



Figure 10: Favourite Betting Strategy by Category

When applying the two basic strategies to the different categorisations, no group appears to perform particularly well or poorly. In Figure 10 the results for the favourite betting strategy are shown. Unlike the accuracy test statistics, no trends appear to emerge. From this strategy, no particular conclusions can be drawn.

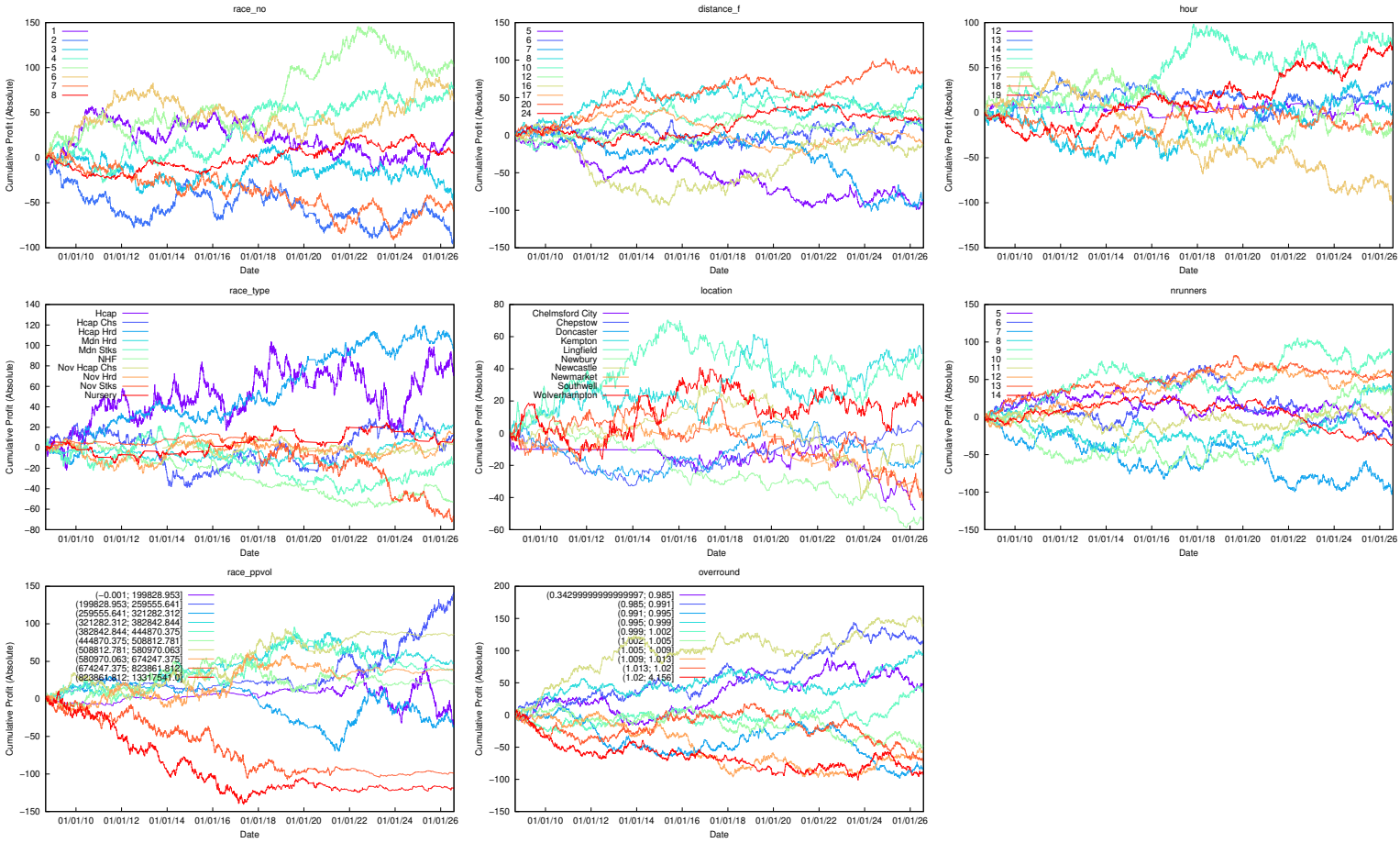


Figure 11: Favourite Hedging Strategy by Category

The favourite hedging strategy is the same as the favourite betting strategy in that no patterns can be found in the results.

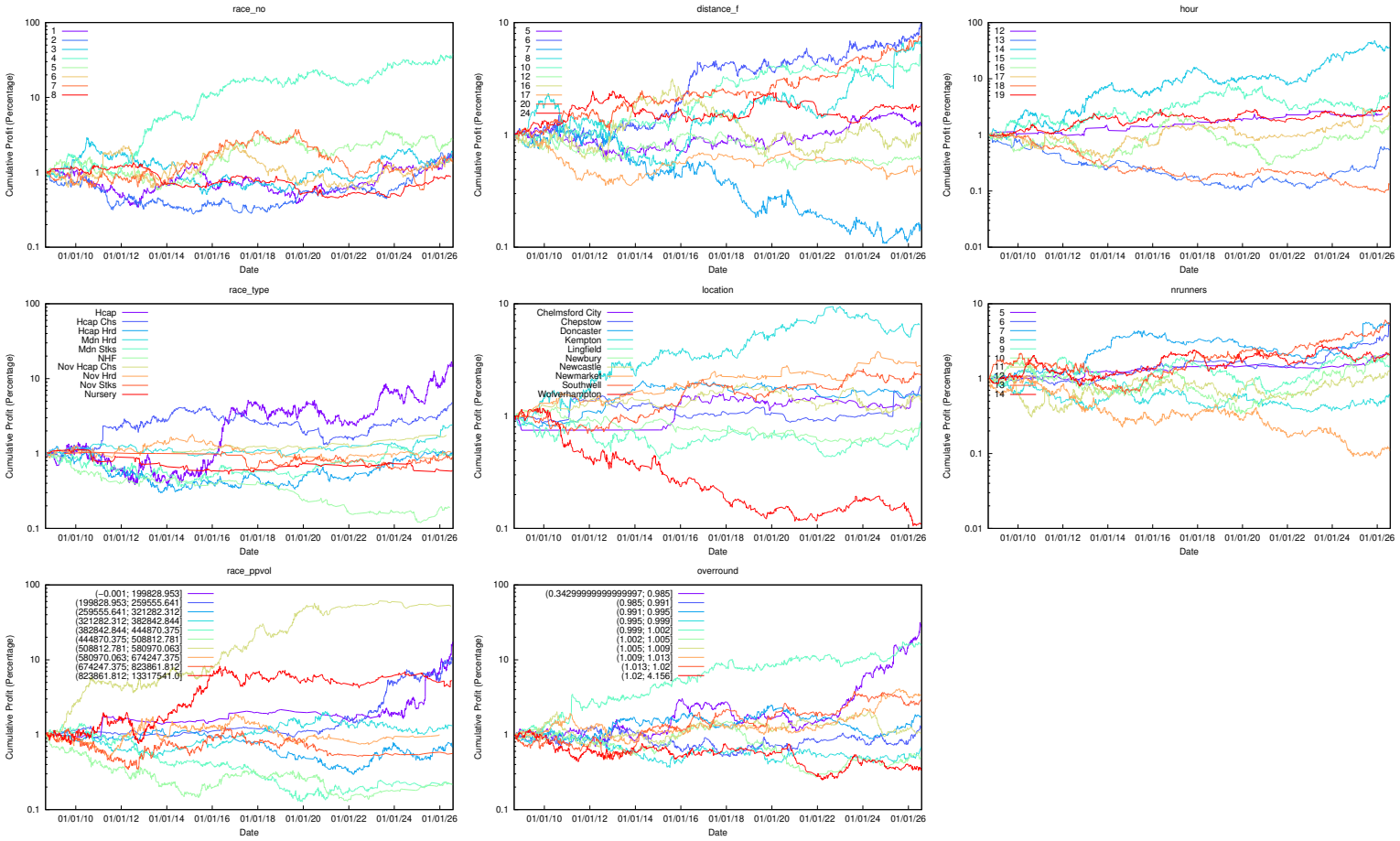


Figure 12: Advanced Betting Strategy by Category

The advanced betting strategy has a wide variation in successfulness. Figure 12 shows the cumulative returns as a multiple of initial wealth. On first inspection, the two most successful implementations of this strategy are for races at 14:00 and races with a total trade volume between £508,813 and £580,970. The least successful was for NHF (National Hunt Flat) races. Again, there does not appear to be much of a link between the different groups and the performance of strategy. The immediate performance of the strategy here is not sufficient to deem any particular grouping of races to have an inefficient market.

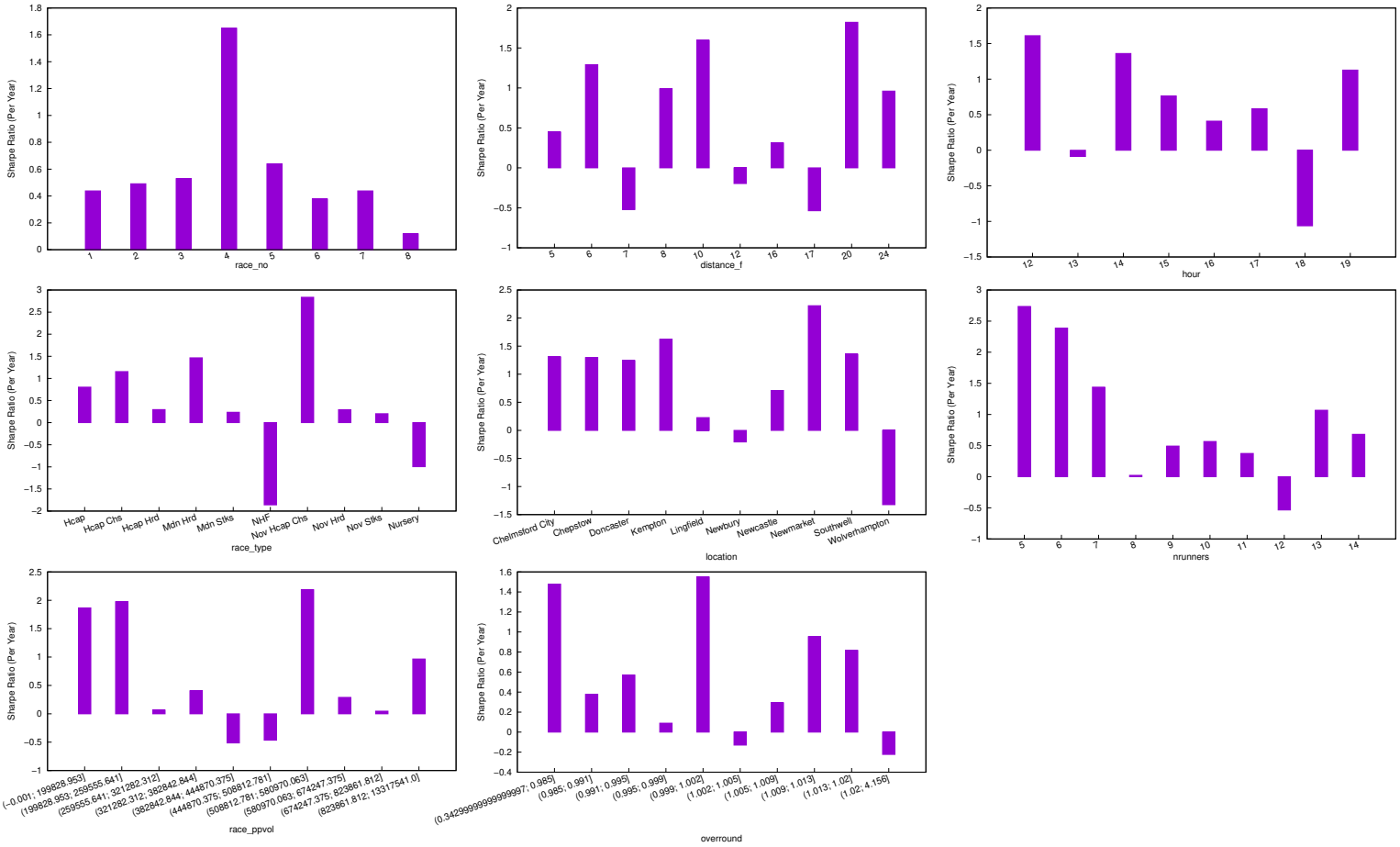


Figure 13: Sharpe Ratio by Category for Advanced Strategy

Sharpe ratios as displayed in Figure 13 help to finalise the picture. It is pretty clear that there is no correlation between the groupings and the performance of the strategy. This is true for all categories except for race number, which appears to follow a similar trend as the results for accuracy (see Figure 8) albeit somewhat exaggerated. While some of the Sharpe ratios are high, I am still reluctant to identify any group as being inefficient. Even with high Sharpe ratios, some groups fail to deliver impressive returns. Races with 5 horses and Novice Handicap Chases have some of the highest Sharpe ratios and yet the actual profit derived from these groups is very limited. Given the variance and lack of correlation, I think it is more likely that these results have come about due to random chance.

6. Legal, Social, Ethical & Professional Issues

6.1. Incentives & Insider Trading

The availability of prediction markets for certain events can incentivise negative acts that otherwise would not be undertaken. At the Super Bowl in 2021, a streaker reportedly bet $50,000 that there would be a streaker (O'Reilly, 2021). The availability of perceived easy money from this bet directly encouraged encouraged a man to engage in illegal behaviour. This was even rationalised (somewhat ironically) as an arbitrage opportunity. Since the cost of streaking is fixed

(ticket costs, transport and fines), any large enough bet placed will result in a guaranteed profit assuming the bettor is able to successfully streak.

In theory, a prediction market could be engineered in such a way as to discourage this. If information was efficiently priced into the market, the possibility of someone going out of their way to streak for market gain would be reflected appropriately in the price. As with all prediction markets, money talks and a large bet on an action signals strong insider information to the market. It is likely that markets for streakers are now more aware of this possibility and are making sure to price it in.

Prediction markets do not just incentivise bettors to take actions that impact the outcome of events. They also incentivise insiders to bet large amounts on outcomes they know beforehand. For the purpose of the market, this may be considered good as it pushes the market towards greater accuracy. In many scenarios however, this information is confidential or sensitive. A U.S. soldier was charged with using classified information about the capture of Nicolás Maduro in Venzuela (U.S. Department of Justice, Office of Public Affairs, 2026). to bet using Polymarket. In the past, sale of state secrets would have to be done face to face with a foreign agent. Now instead, soldiers are incentivised to sell out their country publicly.

Attempts to fix horse races are taken very seriously by the BHA. A 2023 conspiracy to fix a single race by stopping a horse from achieving its best possible place resulted in a 9 year disqualification for the jockey and a 12 year exclusion order for the owner (British Horseracing Authority Judicial Panel, 2025). Investigations and punishments like these enable horse racing markets to operate in an environment that is generally believed to be free from corruption. This is however relevant to a very small portion of highly regulated prediction markets within the UK. When prediction markets are expanded to include all manner of events, specialised investigation boards are impossible to create. It becomes the job of the courts to regulate behaviour, rather than the industry being able to self regulate.

6.2. Gambling

The gambling industry profits from habitual gamblers with addictive personalities. For many people, gambling can become a significant problem. Most gamblers are essentially guaranteed to lose money in the long run, causing financial harm if not managed properly. A bookkeeper fundamentally makes money by preying on peoples' hopes of winning. By engaging with gambling markets, you help to enable problem gambling. Gambling markets are a zero sum game, to succeed you must take money from someone else. By implementing a successful strategy, you are more than likely directly taking money from gamblers with no thought out system or genuine racing model.

Gambling markets look bad when framed this way, but fundamentally they operate extremely similarly to traditional financial markets. Huge amounts of people use the stock market regularly to invest in companies they believe will be successful, but the outcome is essentially the same: a gamble. The primary difference is the social framing of the two. Big established companies and investors are seen as sensible and reasonable while gamblers are considered unreasonable (and usually are). Of course, the engagement that people have with these two markets is very different, even if they operate similarly. When investing in stocks and shares, most people typically invest in a fund curated by an institutional investor. This is not guaranteed to increases in value by any means, but historically has proven to be successful for most average people. If an investor begins to choose individual stocks then they become less likely to succeed. This is

equivalent to typical gambling practices, where the bettor chooses each individual horse to bet on.

Some people would consider a ban on gambling to be a good thing, but it would be safe to say that not many would agree with a ban on purchasing individual stocks and shares. I believe that to be morally consistent, both of these markets should be viewed in the same manner.

6.3. Animal Welfare

Horse races and some other popular prediction markets (such as greyhound racing) rely on the use of live animals. In some cases the animals may become seriously injured or die. Steps have been taken to reduce this number over time. From 2009 to 2025, incidents of race fatality per 1,000 starts fell from 2.00 to 1.07 in the US (The Jockey Club, 2026). In the UK, Aintree racecourse announced in 2013 significant changes to the prestigious Grand National race in the name of improving safety (Scargill, 2023). The maximum number of runners was decreased from 40 to 34. As well as this, multiple course features were altered including the lowering of some fences.

Because of the associated danger, some animal rights groups have attempted to improve conditions for animals or outright ban the events from taking place at all. Any changes imposed may have significant impacts on the markets, as models need to be recalibrated to account for the different environment. Additionally, the availability of the market could be completely jeapordised if events are banned.

7. Conclusion

7.1. Future Work

- compare polling/other forecasting?
- compare betfair tote/bsp values from just bsp bets

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